



Cairo University - Faculty Of Engineering
Computer Engineering Department
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بَلِغ

Literature Review

Baligh: Arabic Writing Assistant

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1 Introduction

Arabic Grammatical Error Correction (GEC) is the most important NLP task when developing writing assistance tools. Arabic has unique challenges because of its morphological richness, orthographic ambiguity, dialectal variations, and limited annotated resources compared to English [Obeid et al., 2020, Habash, 2010]. This document aims to combine the advances in Arabic GEC research to allow us to choose the path for development of our Writing Assistant **Baligh**.

2 Challenges in Arabic NLP

2.1 Morphological Complexity

Arabic exhibits rich inflectional morphology involving gender, number, person, aspect, mood, case, state, and voice, with cliticization ('وصل') of pronouns and particles. MSA verbs can have over 5,400 forms [Obeid et al., 2020]. This complexity significantly impacts GEC system design.

2.2 Orthographic Ambiguity

Arabic is written with optional diacritical marks for short vowels. While native speakers can interpret undiacritized text, this absence is a primary source of ambiguity in ANLP tasks [Obeid et al., 2020].

2.3 Language Variants

Arabic comprises three main versions: Classical Arabic, Modern Standard Arabic (MSA), and regional dialects [Alrehili & Alhothali, 2025]. Each variant requires different handling approaches.

3 Available Datasets and Resources

3.1 GEC Datasets

- **QALB-2014 & QALB-2015:** Primary benchmarks for Arabic GEC, containing native speaker comments from Aljazeera articles [Zaghouani et al., 2015]. These datasets exhibit high punctuation error frequencies (~40% in QALB-2014).
- **ZAEBUC:** Annotated Arabic-English bilingual writer corpus with essays from university students, including spelling/grammar corrections, morphological tokenization, POS tagging, and CEFR ratings [Habash & Palfreyman, 2022].
- **Tibyan Corpus:** A newly developed balanced corpus (~600K tokens) using ChatGPT for data augmentation with expert validation, covering seven error types [Alrehili & Alhothali, 2025].
- **Arabic Learner Corpus (ALC):** Contains L2 learner data with 29 error tags [Alfaifi & Atwell, 2014].

3.2 Treebanks

- **Prague Arabic Dependency Treebank (PADT):** Multi-level linguistic annotations over MSA newswire texts.
- **CAMeL Treebank:** 188K word open-source dependency treebank covering MSA and Classical Arabic across 13 sub-corpora [Habash et al., 2022].

- **I3rab Treebank:** Based on traditional Arabic grammatical theory [Halabi et al., 2020].

"Treebanks are valuable linguistic resources that include the syntactic structure of a language sentence in addition to POS-tags and morphological features. They are mainly utilized in modeling statistical parsers."

— [Halabi et al., 2020]

3.3 Tools

- **CAMeL Tools:** Open-source Python toolkit supporting preprocessing, morphological modeling, dialect identification, NER, and sentiment analysis [Obeid et al., 2020].
- **Farasa:** Fast SVM-based Arabic segmenter achieving state-of-the-art accuracy while being 10x faster than alternatives like MADAMIRA [Abdelali et al., 2016].
- **ARETA:** Automatic error type annotation system achieving 85.8% F1-score, useful for analyzing GEC system outputs [Belkebir & Habash, 2021].

4 Approaches to Arabic GEC

4.1 Rule-Based Methods

Early approaches relied on handwritten linguistic rules and lexicons. Zaghoulani et al. (2015) achieved 66.9% F1 using rules, lexicons, and regular expressions without machine learning. Moukrim et al. (2021) proposed an ontology-based system achieving 92% precision and 84% recall [Moukrim et al., 2021].

4.2 Statistical and Neural Approaches

- **Seq2Seq Transformers:** Solyman et al. (2022) proposed a Transformer-based GEC model with Expectation-Maximization routing to aggregate information across layers, achieving best F1 scores on QALB benchmarks [Solyman et al., 2022]. They also introduced synthetic data generation methods.
- **Pretrained Models:** Alhafni et al. (2023) presented first results using Transformer-based pre-trained Seq2Seq models (fine-tuning CAMeLBERT-MSA) [Alhafni et al., 2023].

Neural approaches might achieve great results but they lack interpretability and require high compute for building and inference.

4.3 Text Editing Approaches

Alhafni & Habash (2025) introduced a text editing approach framing GEC as sequence tagging, where edit tags are assigned to tokens [Alhafni & Habash, 2025].

Key advantages:

- **Efficiency:** Over 6x faster than existing Arabic GEC systems.
- **Language-agnostic:** Derives edit tags directly from data without language-specific rules.
- **SOTA performance** on two Arabic GEC benchmarks.
- **Edit segregation:** Separating punctuation and non-punctuation errors reduces model complexity.

Corrected	النفسيه Alnfsyh	الصحة AlSHh		سيما symA	ولا wLA	بالصحة bAlSHh	الاهتمام AlAhtmAm	يجب yjb	(a)				
	7	6	5	4	3	2	1	0					
Erroneous	النفسيه Alnfsyh	الصحة AlSHh	في fy	ولاسيما wLA symA	لصحه lSHh	ب b	الإهتمام AlAhtmAm	يجب yjb	(b)				
Word Edits	KKKKKKR_[:A_[:]	KKKKR_[:]	DD	KKKI_[]KKKK	MI_[]KKKR_[:]	K	KKR_[]KKKKK	KKK	(c)				
Word Edits (Compressed)	K*R_[:A_[:]	K*R_[:]	D*	KKKI_[]K*	MI_[]K*R_[:]	K*	KKR_[]K*	K*	(d)				
	7b	7a	6b	6a	5	4	3b	3a	2	1b	1a	0	
Tokenized Erroneous	ه## ##h	النفسي Alnfsy	ه## ##h	الصح AlSH	في fy	ولاسيما wLA symA	ه## ##h	لصح lSH	ب b	هتتام## ##htmAm	الإ AlĀ	يجب yjb	(e)
Subword Edits	R_[:A_[:]	KKKKKK	R_[:]	KKKK	DD	KKKI_[]KKKK	R_[:]	MI_[]KKK	K	KKKKK	KKR_[]	KKK	(f)
Subword Edits (Compressed)	R_[:A_[:]	K*	R_[:]	K*	D*	KKKI_[]K*	R_[:]	MI_[]K*	K*	K*	K*R_[]	K*	(g)

Figure 1: Example of Text Editing Tags.

4.4 Grammatical Error Detection (GED)

Alhafni et al. (2023) defined multi-class Arabic GED as a sequence labeling task. A two-stage approach (detect then correct) shows promise for interpretable corrections [Alhafni et al., 2023].

5 Next-Word Prediction

Next-word prediction (NWP) and word auto-completion (WAC) are fundamental NLP tasks that enhance user experience in text input applications by predicting the next word based on context, reducing keystrokes and accelerating text entry [Al-Anzi et al., 2024].

5.1 Statistical Language Models

N-gram language models estimate word probability given previous $n - 1$ words using maximum likelihood estimation. Hamarashid et al. (2021) applied unigram, bigram, and trigram models for Kurdish (morphologically similar to Arabic), demonstrating that statistical approaches remain competitive for resource-limited languages [Hamarashid et al., 2021].

KenLM (Heafield, 2011) is a critical tool for efficient n-gram storage and querying. It implements optimized data structures (PROBING and TRIE) that achieve 2.4x faster queries than SRILM while using 57% of the memory [Heafield, 2011]. KenLM is integrated into major translation systems including Moses and Joshua.

"The count-based models are simple to train, but probabilities of rare n-grams can be poorly estimated due to data sparsity (despite smoothing techniques)."

— [Kim et al., 2016]

5.2 Neural Language Models

Neural Language Models (NLMs) address n-gram data sparsity by learning continuous word representations (embeddings) and using them as inputs to neural networks [Kim et al., 2016]. Key architectures include:

- **LSTM/BiLSTM:** Al-Anzi et al. (2024) achieved 65% accuracy with LSTM and 74% with ARABERT+LSTM on Arabic word prediction [Al-Anzi et al., 2024].
- **Medical Context:** Habib et al. (2021) found BiLSTM with 4-gram context outperformed other configurations for Arabic medical text [Habib et al., 2021].

5.3 Pre-trained Models

- **Aravec**: Pre-trained Arabic word embeddings (300-dim) from 3 billion words of Twitter and Wikipedia data using Skip-gram [Habib et al., 2021].
- **ARABERT**: Arabic BERT providing contextualized embeddings. Al-Anzi et al. (2024) demonstrated ARABERT preprocessing improves accuracy by $\sim 10\%$ over NLTK processing and reduces unique tokens from 47,820 to 10,375 [Al-Anzi et al., 2024].
- **BERT vs. CBOW**: Taher (2024) compared BERT and Continuous Bag of Words (CBOW) for Arabic, finding that while CBOW achieved higher accuracy on closed datasets, BERT provided more robust context understanding for open-domain prediction [Taher, 2024].

5.4 Word-Level Performance Comparison

The following table illustrates the performance of various architectures on Arabic next-word prediction tasks [Al-Anzi et al., 2024]:

Model	Training Accuracy	Validation Accuracy
Plain LSTM	65%	63%
ARABERT + LSTM	74%	88%
Markov Model	78%	77%

Table 1: Arabic Next-Word Prediction Results.

5.5 Character-Level Approaches

Kim et al. (2016) proposed replacing word embeddings with character-level CNN outputs for language modeling [Kim et al., 2016]. The architecture uses temporal convolutions over character sequences, max-over-time pooling, and highway networks to produce word representations.

Key benefits for morphologically rich languages like Arabic:

- Reduces vocabulary from 100K+ words to ~ 40 -50 characters.
- Naturally handles out-of-vocabulary words through character composition.
- Captures morphological patterns including Arabic root structures (جذر).
- Achieved SOTA perplexity on Arabic (136.61) with fewer parameters at the time (2016).

"While NLMs have been shown to outperform count-based n -gram language models... they are blind to subword information (e.g. morphemes). For example, they do not know, a priori, that eventful, eventfully, uneventful, and uneventfully should have structurally related embeddings in the vector space."

— [Kim et al., 2016]

5.6 Advanced Generation & Decoding

- **Transformer-Based Generation: AraGPT2** (Antoun et al., 2021) represents the state-of-the-art in pure generation, trained on large Arabic corpora to handle complex morphology [Antoun et al., 2021].

- **Constrained Decoding:** To ensure grammatical correctness in suggestions, **NeuroLogic Decoding** (Lu et al., 2021) introduces a method to satisfy logical constraints (e.g., required parts of speech) during the generation process [Lu et al., 2021].

6 Writing Assistant Systems

6.1 ARWI (Arabic Write and Improve)

Chirkunov et al. (2025) presented ARWI, a comprehensive Arabic writing assistant featuring:

- Prompt database for different proficiency levels
- Arabic text editor
- Grammatical error detection and correction
- Automated essay scoring aligned with CEFR standards (A1-C2)

The system fine-tunes CAMELBERT-MSA and provides actionable feedback for learners [Chirkunov et al., 2025].

7 Supporting Technologies

- **Diacritization:** AlKhamissi et al. (2020) proposed a hierarchical recurrence architecture for Arabic diacritization achieving 5.34% WER [AlKhamissi et al., 2020].
- **Language Models:** KenLM (Heafield, 2011) provides efficient n-gram language model queries useful for next-word suggestion and fluency scoring [Heafield, 2011].
- **Morphological Analysis:** Parsing models supported by morphological analysis achieve higher accuracy when sentences are diacritized [Almunirawi & Baraka, 2021].

8 Error Taxonomy

Based on ARETA [Belkebir & Habash, 2021] and ALC [Alfaifi & Atwell, 2014], Arabic errors are categorized into:

Class	Err Tag	Description	Arabic Example	Transliteration
Orthography	OA	Confusion in Alif, Ya and Alif-Maqsurā	على ← على	çly → çlȳ
	OC	Wrong order of word characters	تبرينا ← تبرينا	tbrynA → trbynA
	OD	Additional character(s)	يعنوم ← يعنوم	yçdwm → ydwm
	OG	Lengthening short vowels	نقيم ← نقيم	nqymw → nqym
	OH	Hamza errors	أكثر ← أكثر	Ak0r → Āk0r
	OM	Missing character(s)	سألين ← سألين	sAlyn → sAȳlyn
	ON	Confusion between Nun and Tanwin	ثوين ← ثوين	0wbn → 0wbū
	OR	Replacement in word character(s)	مصلنا ← وصلنا	mSlnA → wSlnA
	OS	Shortening long vowels	أوقت ← أوقت	Āwqt → ĀwqAt
	OT	Confusion in Ha, Ta and Ta-Marbuta	مشاركه ← مشاركة	mšArkḥ → mšArkḥ
	OW	Confusion in Alif Fariqa	وكانوا ← وكانوا	wkAnw → wkAnwA
	OO	<i>Other orthographic errors</i>	-	-
Morphology	MI	Word inflection	معرّوف ← عارف	mçrwf → çArf
	MT	Verb tense	تفرحنى ← أفرحتني	tfrHny → ĀfirHny
	MO	<i>Other morphological errors</i>	-	-
Syntax	XC	Case	رائع ← رائعا	rAȳç → rAȳçAā
	XF	Definiteness	السن ← سن	Alsn → sn
	XG	Gender	الغربي ← الغربية	Alȳrby → Alȳrbyḥ
	XM	Missing word	على ← Null	Null → çlȳ
	XN	Number	فكرتي ← افكاري	fkrty → ĀfkAry
	XT	Unnecessary word	على ← Null	çlȳ → Null
	XO	<i>Other syntactic errors</i>	-	-
Semantics	SF	Conjunction error	سبحان ← فسبحان	sbHAn → fsbHAn
	SW	Word selection error	من ← عن	mn → çn
	SO	<i>Other semantic errors</i>	-	-
Punctuation	PC	Punctuation confusion	المتوسط ← المتوسط	AlmtwsT. → AlmtwsT,
	PM	Missing punctuation	العظيم ← العظيم،	AlçDym → AlçDym,
	PT	Unnecessary punctuation	العام. ← العام	AlçAm, → AlçAm
	PO	<i>Other errors in punctuation</i>	-	-
Merge	MG	Words are merged	ذهبت اليارحة ← ذهبت اليارحة	0hbtAlbArHh → 0hbt AlbArHh
Split	SP	Words are split	المحادثات ← المحادثات	AlmHA d0At → AlmHAD0At

Table 1: The ALC error type taxonomy extended with merge and split classes. The error tags are listed alphabetically, except for the highlighted **Other* tags, which we do not support in ARETA.

Figure 2: Taxonomy of Common Arabic Grammatical Errors

- **Orthographic:** Spelling, Hamza, Ta-Marbuta errors
- **Morphological:** Word formation, inflection errors
- **Syntactic:** Agreement, case marking, word order
- **Semantic:** Lexical choice errors
- **Punctuation:** Missing/incorrect punctuation
- **Merge/Split:** Word boundary errors

9 Key Findings and Gaps

- **Best Performance:** Text editing approaches achieve SOTA while being 6x faster [Alhafni & Habash, 2025].
- **Multi-task Learning:** GED+GEC joint approaches improve overall performance.
- **Research Gaps:**
 1. Limited resources for dialectal Arabic GEC.
 2. Need for real-time writing assistance tools.
 3. Lack of integration between GEC, diacritization, and next-word prediction.

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